

INTERNSHIP REPORT

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degree of

BACHELOR OF TECHNOLOGY

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by

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INDEX

S. No.	Content	Page No.
1.	Organization Information	3
2.	Technology Currently Working With	4
3.	Project Introduction	7
4.	Progress done till presentation	7
5.	My Learnings	8
6.	Work yet to complete	8
7.	Conclusion	9

Organization Information

Cognizant Technology Solutions is a leading American multinational provider of information technology (IT) services, consulting, and business process outsourcing (BPO) solutions. Although headquartered in Teaneck, New Jersey, USA, the company has deep roots in India, where it was originally established in 1994 as Dun & Bradstreet Satyam Software (DBSS), an in-house technology unit of Dun & Bradstreet.

Cognizant began its journey in Chennai with a team of 50 employees focused on delivering largescale IT projects for Dun & Bradstreet. By 1996, the company expanded its focus beyond internal clients and began serving external customers. That same year, Dun & Bradstreet spun off multiple subsidiaries—including DBSS—to form Cognizant Corporation. In 1997, DBSS was renamed Cognizant Technology Solutions, marking the official emergence of the brand recognized today.

Over the years, the organization has evolved into one of the world's most respected professional services firms, serving major clients across industries such as financial services, healthcare, retail, communications, life sciences, and manufacturing with more than 250,000 associates.

The Quality Engineering & Assurance (QEA) service line focuses on ensuring the quality and reliability of enterprise processes, applications, and systems through intelligent automation, AI-driven testing, and modern assurance practices. QEA accelerates business transformation by embedding quality throughout the development lifecycle, reducing manual efforts, ensuring regulatory compliance, and enabling faster time-to-market.

Cognizant's QEA practice emphasizes shift-left testing, continuous testing in CI/CD pipelines, and leveraging AI/ML for test optimization, making it a leader in modern quality engineering practices

Technology Currently Working With

I am undergoing training in the Generation Cognizant (GenC) Quality Engineer with Selenium Automation program (Java Track), consisting of two stages followed by business-aligned projects.

Stage 1: Core Java with SQL (Days 1–21)

Core Java Fundamentals

- Learned **Object-Oriented Programming (OOP)** concepts such as encapsulation, inheritance, polymorphism, and abstraction.
- Practiced writing Java programs using **classes, objects, constructors, and methods**.
- Studied **exception handling** using try-catch, finally blocks, and custom exceptions to handle runtime errors.
- Explored the **Java Collections Framework** including ArrayList, HashMap, HashSet, LinkedList, and TreeSet for efficient data management.
- Learned **Java Streams and functional programming concepts** for processing collections using filter, map, and reduce operations.
- Studied **multithreading concepts** to understand concurrent execution and thread lifecycle.
- Practiced coding exercises and hands-on assignments using **Eclipse IDE and JDK 17**.

SQL and Database Concepts

- Learned **Data Definition Language (DDL)** commands such as CREATE, ALTER, and DROP to manage database structures.
- Practiced **Data Manipulation Language (DML)** commands including INSERT, UPDATE, DELETE, and SELECT to manipulate database records.
- Worked with **joins (INNER JOIN, LEFT JOIN, RIGHT JOIN)** to combine data from multiple tables.
- Used **subqueries and nested queries** to retrieve complex data sets.
- Studied **database normalization techniques** to eliminate redundancy and maintain data integrity.
- Learned **indexing concepts** to improve database query performance.

- Executed SQL queries and database operations using **MySQL tools**.

Development Tools Used

- Eclipse IDE for Java programming and debugging.
- JDK 17 for compiling and running Java applications.
- MySQL database for storing and querying structured data.

Stage 2: Functional Testing with Selenium Automation (Days 22–65)

Functional Testing Concepts

- Studied the **Software Development Life Cycle (SDLC)** phases such as requirement analysis, development, testing, deployment, and maintenance.
- Learned the **Software Testing Life Cycle (STLC)** including test planning, test case creation, test execution, defect tracking, and test closure.
- Designed **test cases and test scenarios** based on application requirements.
- Understood the **defect lifecycle**, including defect identification, reporting, verification, and closure.
- Learned different types of testing such as **functional testing, regression testing, smoke testing, and exploratory testing**.

Selenium WebDriver

- Learned the architecture and setup of **Selenium WebDriver for automation testing**.
- Used **locators (ID, Name, XPath, CSS Selector)** to identify elements on web pages.
- Implemented **synchronization techniques** such as implicit and explicit waits.
- Practiced handling **alerts, frames, windows, and dropdown menus** in automated tests.
- Implemented the **Page Object Model (POM)** design pattern for maintainable test scripts.

Test Automation Frameworks

- Used **TestNG framework** to organize test cases and generate test reports.
- Configured test execution using **TestNG XML and JSON files**.
- Implemented **data-driven testing** using Apache POI to read test data from Excel files.

Advanced Automation Techniques

- Created **dynamic XPath expressions** to locate complex web elements.
- Used **JavaScript Executor** to perform advanced browser interactions.

- Implemented the **Actions class** for mouse and keyboard actions such as drag-and-drop and hover.

DevOps Integration

- Used **Git** for version control and source code management.
- Learned **Maven** for project dependency management and build automation.
- Integrated automation scripts with **Jenkins** for CI/CD pipelines.
- Used **Selenium Grid** for distributed test execution across multiple browsers and environments.
- Learned basic **Docker** containerization concepts for test environment setup.

Behavior Driven Development (BDD)

- Implemented **Cucumber framework** for writing test scenarios in Gherkin language.
- Connected feature files with step definitions for automated execution.

API Testing

- Used **Postman** for testing REST APIs and validating responses.
- Implemented API automation using **REST Assured framework**.

Test Reporting

- Generated detailed automation test reports using **ExtentReports** and **Allure Reports**.

Additional Skills Developed

- Understanding of **Agile and Scrum methodologies** used in modern software development.
- Experience with **Jira tool** for defect tracking and project management.
- Introduction to **Generative AI fundamentals** for improving testing efficiency.
- Understanding **Continuous Integration and Continuous Deployment (CI/CD)** pipelines.
- Learning to design **hybrid automation frameworks combining multiple testing approaches**.

Project Introduction

I have not been assigned any project till now . Currently I am going under training. Project will be assigned in next week . So currently I have nothing to show under project work. I am working under the role of an intern.

Progress Done Till Presentation

Internship start date: 31st January 2026

Internship Joining Location: Cognizant Technology Solutions, Chennai , India

From 31st Jan to 5th Feb 2026:

phase 1: Icebreaker (Days 1-5)

- Corporate induction, core values, leader talks
- Behavioral training, talent management connect[file:2]

From 6th Feb to 9th March 2026:

Phase 2: Stage 1 - Core Java & SQL (Days 6-21)

Days 1-15: Core Java (Udemy + Tekstac hands-on + Eclipse practice)

- OOP, control structures, arrays, strings, collections, exceptions, streams
- Completed 100% hands-on (Streamline Health, Water Tank, etc.)
- ICT-Java Assessment, Code Challenges

Days 16-19: SQL fundamentals, joins, subqueries

Day 20-21: Stage 1 Assessment

From 10th March to 13th March 2026:

Phase 2: Stage 2 –Functional testing (Days 22-30)

completed 100% Stage 1 hands-on; progressing through Stage 2 with daily code challenges and project work.

My Learnings & Work Yet to Complete

Remaining Stage 2 Training (Days 37-65):

- Days 37-39: POM + Apache POI for data-driven testing
- Days 40-45: TestNG advanced (parallel execution), Interim Project Evaluation
- Days 46-54: Automation Framework, DevOps (Maven, Grid, Jenkins)
- Days 55-56: BDD Cucumber
- Days 57-61: Postman + REST Assured API automation
- Days 62: Docker/Containerization
- Days 63-65: Final Project Evaluation + Technical Interview[file:2]

Project Milestones:

- Complete Mini Project: Hybrid framework with CI/CD, reporting
- Hackathon: Group project with full stack (Selenium + API + DevOps)
- All project deliverables: Source code, execution reports, demos
- Prepare for Final Technical Evaluation (video interview with BU SME)

CONCLUSION

This internship has provided foundational training in modern Quality Engineering practices through Cognizant's GenC QEA program. Progressing from Core Java/SQL foundations to advanced Selenium automation and DevOps integration, I've gained practical skills applicable to real-world testing scenarios.

Key achievements include completing Stage 1 with 100% hands-on, mastering WebDriver/POM, and initiating project automation. The flipped classroom approach fostered self-learning while SME guidance accelerated technical growth.

Moving forward, I am committed to completing Stage 2 modules, delivering all project artifacts, and excelling in evaluations. This experience positions me as a capable Quality Engineer ready for production environments, contributing to Cognizant's AI-driven quality transformation initiatives.

my internship so far, I have developed a comprehensive understanding of full-stack development using the MEAN stack. I gained proficiency in Angular for building dynamic, single-page applications and used Bootstrap to ensure responsive, mobile-first user interfaces. On the server side, I learned to build scalable APIs with Node.js and Express.js, while mastering MongoDB for flexible, documentoriented database management.

Beyond coding, I have integrated AWS into my workflow to understand cloud infrastructure and deployment. I also gained hands-on experience with Jenkins, learning how to automate build and deployment processes through Continuous Integration and Continuous Deployment (CI/CD) pipelines. Combined with Git for version control and Agile/Jira for project management, these skills have allowed me to understand the full lifecycle of a modern web application.